Advanced Programming Practice · 21CSC203P

Unit 1 · Session 4 · SLO 2 — Solved Worksheet

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Unit – 1 : Session – 4 : SLO – 2 | Multi-Paradigm Use

■ Activity (Solutions)

• 1) OOP + functional: class Greeter { String prefix; Greeter(String p){ prefix=p; } } java.util.function.Function<String,String> f = name -> "Hello " + name; System.out.println(f.apply("Sam")); — lambda used alongside a class.

• 2) Java is multi-paradigm because it natively supports object-oriented constructs (classes, interfaces, inheritance), imperative/procedural constructs (loops, static methods, mutable state) and functional constructs (lambdas, streams, Function/Supplier/Consumer interfaces) in one language.

• 3) Procedural → OO conversion: Procedural: static void printArea(int l, int w){ System.out.println(l*w); } OO: class Rectangle { int l, w; Rectangle(int l, int w){ this.l=l; this.w=w; } int area(){ return l*w; } void print(){ System.out.println(area()); } } new Rectangle(4,5).print(); — data and behavior are encapsulated in the object.